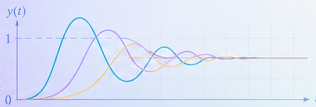
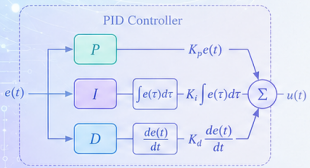
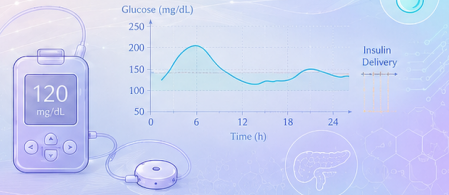
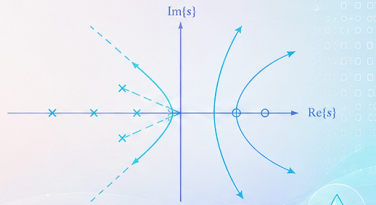
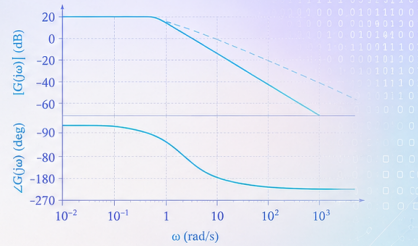


Introduction

SAHAJ SAXENA
sahajsaxena.github.io



$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



Learning Objectives

By the end of this lecture, students should be able to:

- ▶ define a **control system** and explain its importance in engineering;
- ▶ distinguish between **open-loop** and **closed-loop control systems**;
- ▶ identify the basic components of a **feedback control system**;
- ▶ explain the role of feedback in improving system performance;
- ▶ classify control systems using common engineering criteria;
- ▶ identify real-world applications of control systems in electrical, computer, and interdisciplinary engineering.

Prerequisites: Basic knowledge of differential equations, Laplace transforms, electrical circuits, and elementary system modelling.

Evaluation Scheme

1. Mid Semester Test 30 Marks
2. End Semester Test 40 Marks
3. Session 30 Marks

Sessional includes:

- ▶ Quiz 10 Marks (28th August 2026)
- ▶ Lab Quiz 10 Marks
- ▶ Lab Performance 5 Marks
- ▶ Lab Record 5 Marks

Lecture notes will be posted on webpage at sahaj.saxena@github.io

Why Study Control Systems?

Many engineering systems must operate **automatically** despite

- ▶ disturbances and environmental changes;
- ▶ uncertain system parameters;
- ▶ measurement noise;
- ▶ changing operating conditions;
- ▶ safety and performance constraints.

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Central Question

How can we make a physical or computational system behave in a **desired, accurate and reliable** manner?

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A control system continuously determines the required input so that the system output follows the desired behaviour.

Control objective: $y(t) \rightarrow r(t)$

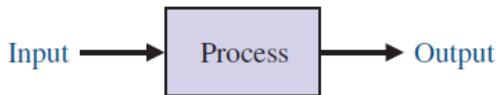
where $r(t)$ is the desired reference and $y(t)$ is the measured output.

Basic Terminology

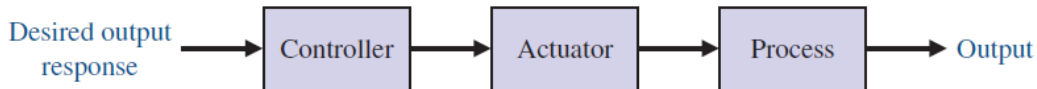
- ▶ Control engineers are concerned with understanding and controlling segments of their environment, often called **systems**, which are interconnections of elements and devices for a desired purpose.
- ▶ A control system consists of interconnected components to achieve a desired purpose.
- ▶ Control engineering deals with the design (and implementation) of control systems using **mathematical models** representing actual physical nonlinear, time-varying systems with parameter uncertainties in the presence of external disturbances.
- ▶ A **sensor** is a device that provides a measurement of a desired external signal. For example, resistance temperature detectors (RTDs) are sensors used to measure temperature.
- ▶ An **actuator** is a device employed by the control system to alter or adjust the environment. An electric motor drive used to rotate a robotic manipulator is an example of a device transforming electric energy to mechanical torque.

Open-loop control

- ▶ A component, or process, to be controlled can be represented graphically



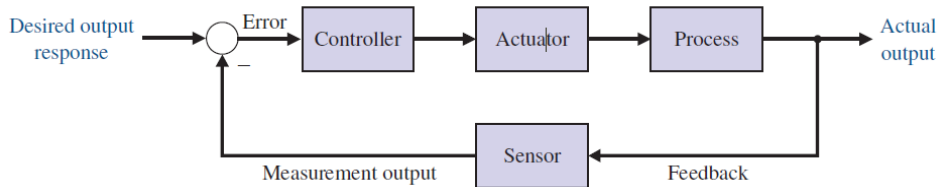
- ▶ An open-loop control system uses a controller and an actuator to obtain the desired response



- ▶ An open-loop control system utilizes an actuating device to control the process directly **without using feedback**.

Closed-loop control aka feedback system

- ▶ A closed-loop control system utilizes an additional measure of the actual output to compare the actual output with the desired output response. The measure of the output is called the **feedback** signal.

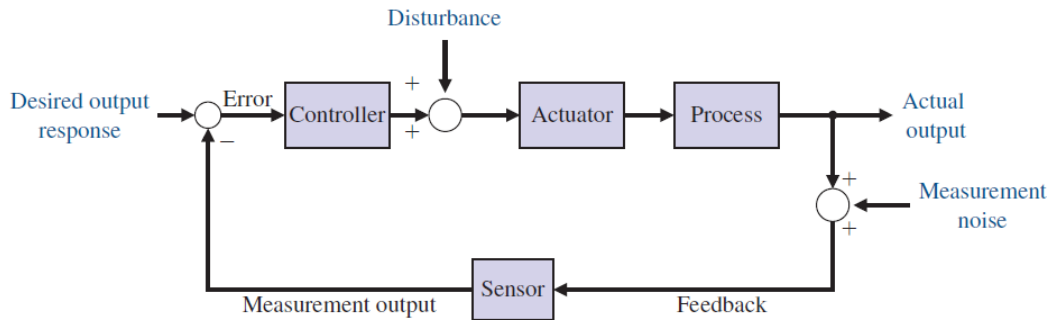


- ▶ A closed-loop control system uses a measurement of the output and feedback of this signal to compare it with the desired output (reference or command) and use the **difference as a means of control**.
- ▶ With an accurate sensor, the measured output is a good approximation of the actual output of the system.

Advantages of closed-loop system

A closed-loop control has many advantages over open-loop control, including

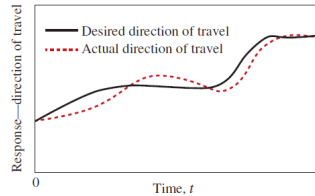
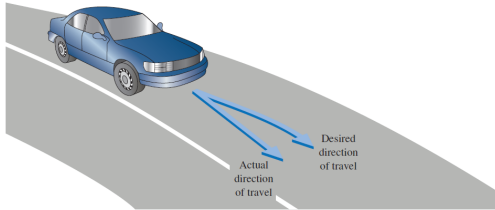
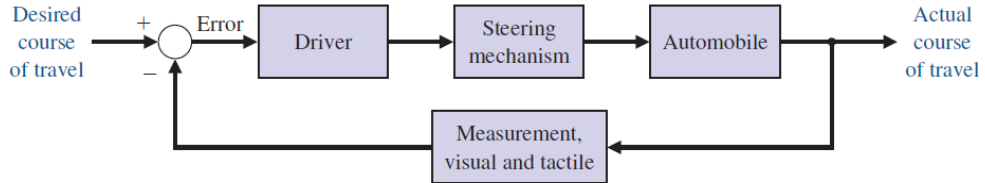
- ▶ the ability to reject external disturbances
- ▶ improve measurement noise attenuation.



Examples

- ▶ A common example of an open-loop control system is a microwave oven set to operate for a fixed time.
- ▶ An example of a closed-loop control system is a person steering an automobile (assuming his or her eyes are open) by looking at the auto's location on the road and making the appropriate adjustments.

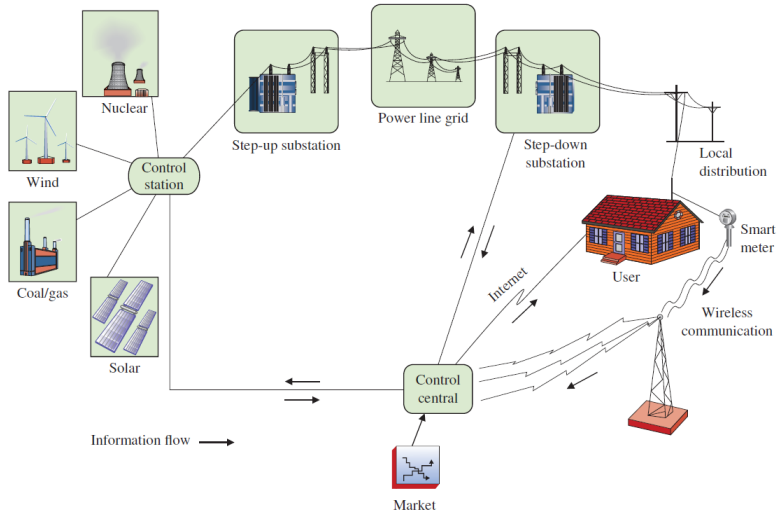
Automated Vehicles



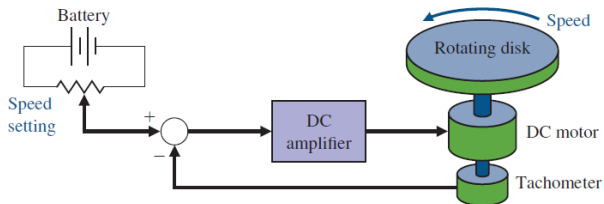
Autonomous systems



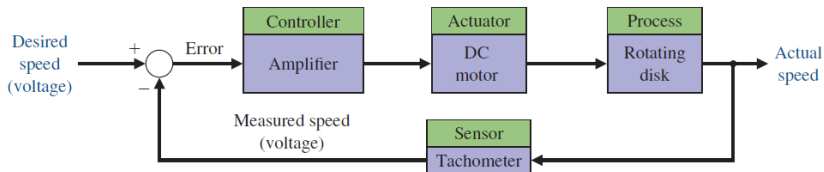
Smart grids—distribution networks that measure and control usage



(a) CL control of speed of rotating disk. (b) Block diagram model

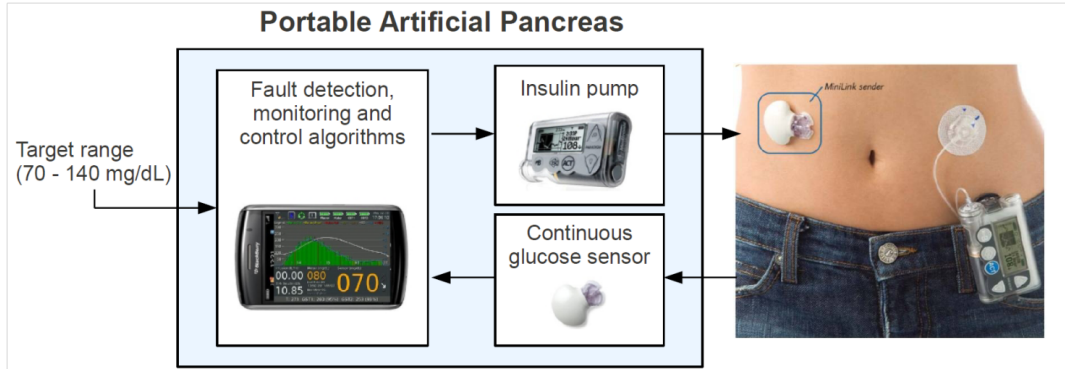


(a)



(b)

Insulin delivery control system



Control is everywhere



Recommended Textbooks and References

Primary Textbooks

1. Richard C. Dorf and Robert H. Bishop

Modern Control Systems

Pearson.

2. Norman S. Nise

Control Systems Engineering

Wiley.

3. Katsuhiko Ogata

Modern Control Engineering

Pearson.

Recommended approach: Use Nise or Dorf as the primary text and consult Ogata for additional solved examples.

Additional References

4. Benjamin C. Kuo and Farid Golnaraghi

Automatic Control Systems

Wiley.

5. Gene F. Franklin, J. David Powell and Abbas Emami-Naeini

Feedback Control of Dynamic Systems

Pearson.

6. M. Gopal

Control Systems: Principles and Design

McGraw Hill.

Thank You

Questions and Discussion